CRICKSTATS

Cricket Player Performance Tracking Application

Salesforce Administrator with AI Agentforce

Project Team

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Demo Video: Watch the CRICKSTATS demo video

1. Project Overview

Cricket teams often keep player and match records in spreadsheets or on paper. This makes it hard to get accurate statistics, compare players and pick teams. CRICKSTATS moves this data into Salesforce so that:

- Player details are stored in one central place.
- Match-by-match performance is recorded against each player.
- Totals and averages are calculated automatically.
- Users can look up players by team from the app home page.

2. Problem Statement

Who	Wants to	But (Problem)	Because	Feels
Cricket team coach	Track player performance across matches	Cannot get accurate statistics easily	Data is stored in different files and formats	Frustrated
Team manager	Maintain complete player records	Updating and verifying data takes too long	Records are maintained manually	Stressed
Sports analyst	Analyze player trends & generate reports	Comparing performances is difficult	There is no centralized analytics system	Confused

3. Solution Architecture

Layer	Technology
Presentation	Salesforce Lightning App (CRICKSTATS)
Application logic	Screen Flow, Formula field, Roll-up Summary fields
Data	Custom objects: Player and Match Performance
Relationship	Master-Detail (Player is parent, Match Performance is child)

Data Model & Flow

```
      Data flow: User -> Lightning App -> Player object <- Match Performance object
-> Roll-up Summary -> Player Flow (Home page) Relationship: Player (parent) 1 -----
* Match Performance (child) (Master-Detail relationship through the "Player" field
on Match Performance)
```

Because the relationship is Master-Detail, roll-up summary fields on Player automatically summarize the Match Performance records of that player.

4. Implementation

Milestone 1: Salesforce Developer Account

Activity 1 & 2: Creating and Activating the Developer Account

1. Opened the Salesforce developer signup page. Entered details (Role: Developer, Company: College).
2. Submitted the form, opened the verification email, and set a password.
3. Successfully redirected to the Salesforce Setup page of the Developer Edition org.

Milestone 2: Object Creation (Part 1)

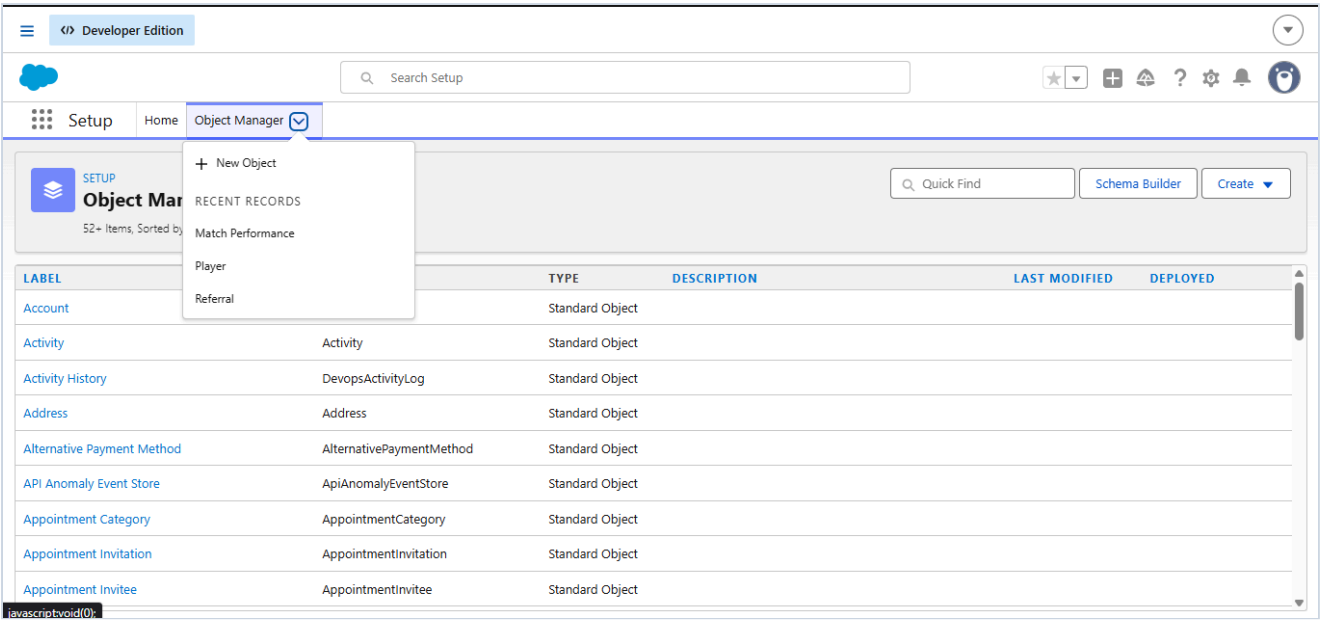
Activity 1: Player object (Setup > Object Manager > Create > Custom Object)

Setting	Value
Label	Player
Plural Label	Players
Record Name	Player Name (Text)
Optional features	Allow Reports, Allow Activities, Track Field History

Milestone 2: Object Creation (Part 2)

Activity 2: Match Performance object

Setting	Value
Label	Match Performance
Plural Label	Match Performances
Record Name	Match Performance Name
Data Type	Auto Number (Format: MP-{0000}, Starting Number: 1)
Optional features	Allow Reports, Allow Activities



Milestone 3: Custom Tabs

Path: Setup > Tabs > Custom Object Tabs > New

1. Created a tab for the **Player** object and chose a tab style.
2. Kept the default profile visibility.
3. Repeated the steps to create a tab for the **Match Performance** object.

Milestone 4: The Lightning App

Path: Setup > App Manager > New Lightning App

Setting	Value
App Name	CRICKSTATS
Navigation style	Standard navigation

Milestone 5: Fields and Relationships (Player Object)

Custom fields were added to capture detailed information about each player.

Field Label	Data Type	Notes
Player Name	Text	Standard record name field
Player Number	Text (3)	Jersey number
Profile Picture	Text Area (Rich)	Player photo
Date of Birth	Date	
Role	Picklist	Batsman, Bowler, All Rounder, Wicket Keeper
Batting Style	Text (50)	Example: Right-Handed
Bowling Style	Text (50)	Example: Right-Arm Fast
Nationality	Picklist	Australia, India, England, etc.
Team	Picklist	IPL franchises
Match Performance	Lookup Relationship	Lookup to Match Performance
AVG Score	Formula (Number)	Total Runs divided by Matches Played

AVG Score Formula Calculation:

```
IF( Matches_Played__c > 0, Total_Runs__c / Matches_Played__c, 0 )
```

Milestone 5: Fields and Relationships (Match Performance Object)

Custom fields designed to record per-match statistics for a player.

Field Label	Data Type
Match Performance Name	Auto Number (MP-{0000})
Player	Master-Detail Relationship (Player)
Match Date	Date
Opponent Team	Text (100)
Match Result	Picklist (WON, LOST, DRAW)
Player of the Match	Checkbox
Runs Scored	Number
Balls Faced	Number
Fours	Number
Sixes	Number
Overs Bowled	Number
Wickets Taken	Number
Runs Conceded	Number
Maiden Overs	Number
Stumpings	Number
Catches	Number

Note on the Master-Detail relationship: The Player field was initially created as a Lookup. Salesforce restricts changing a Lookup to Master-Detail if records exist or under certain conditions in a fresh org. To resolve this, the Lookup field was deleted and recreated as a Master-Detail relationship to Player. This is a mandatory step to enable roll-up summary fields on the Player object.

Milestone 6: Roll-up Summary Fields

Created on the **Player** object (the master side) to aggregate data from **Match Performances**.

Field Label	Roll-up Type	Field Aggregated	Filter Criteria
Matches Played	COUNT	-	All records
Number of 50's	COUNT	-	Runs Scored >= 50 AND < 100
Number of 100's	COUNT	-	Runs Scored >= 100
Total 4's	SUM	Fours	All records
Total 6's	SUM	Sixes	All records
Total Runs	SUM	Runs Scored	All records
Total Overs	SUM	Overs Bowled	All records
Total Wickets	SUM	Wickets Taken	All records
Total Stumpings	SUM	Stumpings	All records
Total Catches	SUM	Catches	All records

Milestone 7: Flows

Activity 1: Screen Flow (Player Flow)

Path: Setup > Flows > New Flow > Screen Flow

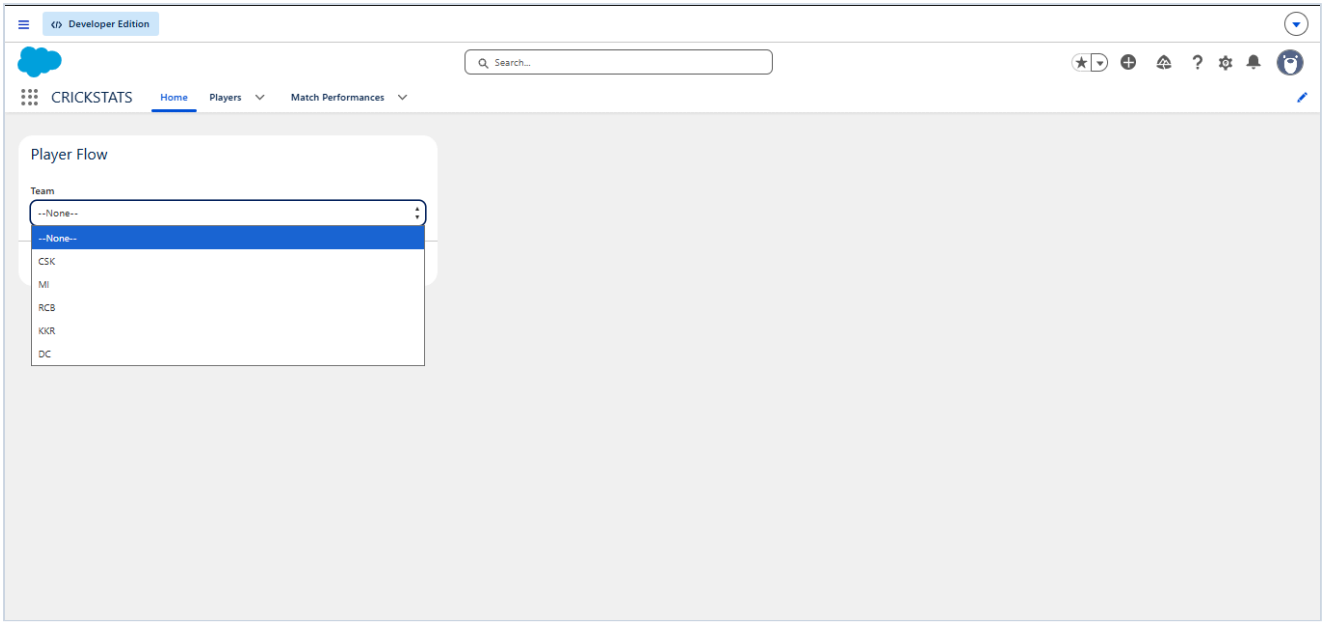
- **Element 1: Screen "Players screen"**
  Added a Picklist component labeled **Team** (API: `Team_Select`). Uses a Picklist Choice Set resource (`TeamChoiceSet`) mapped to Player > Team.
- **Element 2: Get Records "Players"**
  Filters the Player object where `Team` Equals the dynamically selected value from the screen. Stores all matching records.

- **Element 3: Screen "Player_info"**
  Displays a **Data Table** component containing the sourced collection. Columns included: Player Name, Profile Picture, Total Runs, AVG Score, Batting Style, Bowling Style, Matches Played, Number of 50's, Number of 100's.

Saved as Active Flow with API Name: `Player_Flow`.

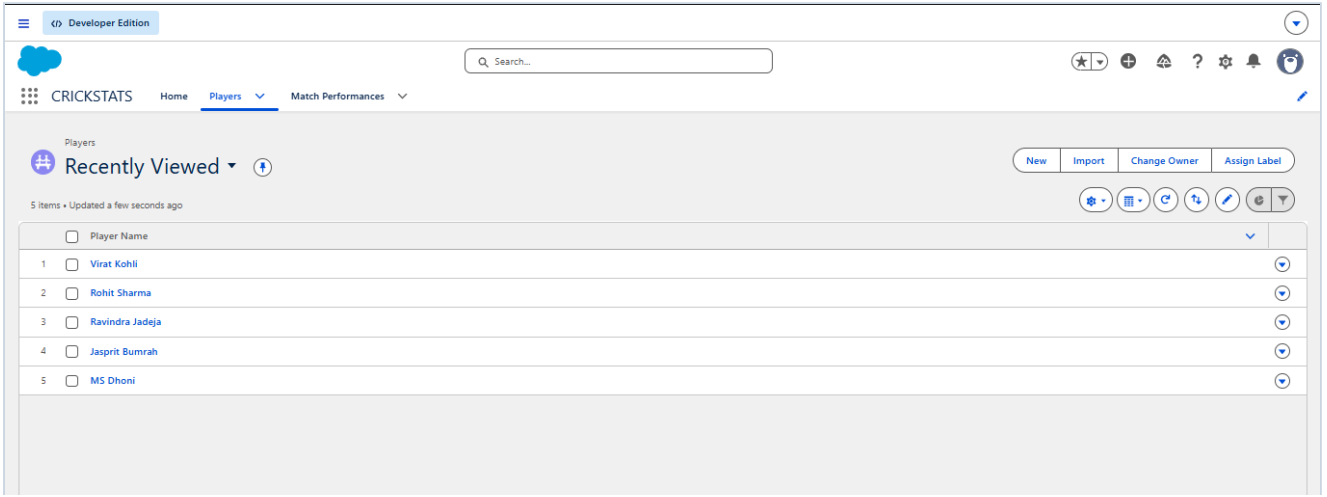
Activity 2: Flow on the Home Page

1. Added the **Home** item to the CRICKSTATS app navigation bar.
2. Opened Lightning App Builder via **Edit Page** on the Home screen.
3. Removed default components and added a **Flow** component, pointing to **Player Flow**.
4. Activated and assigned as the App Default.

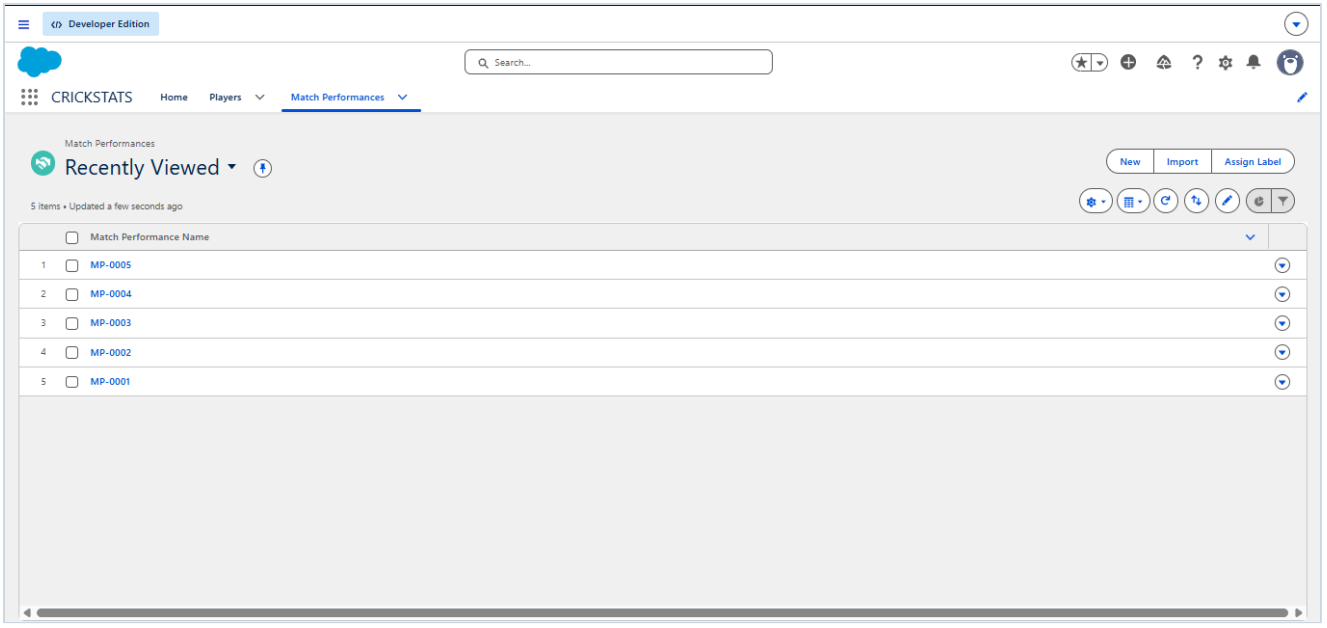


5. Sample Data and Testing

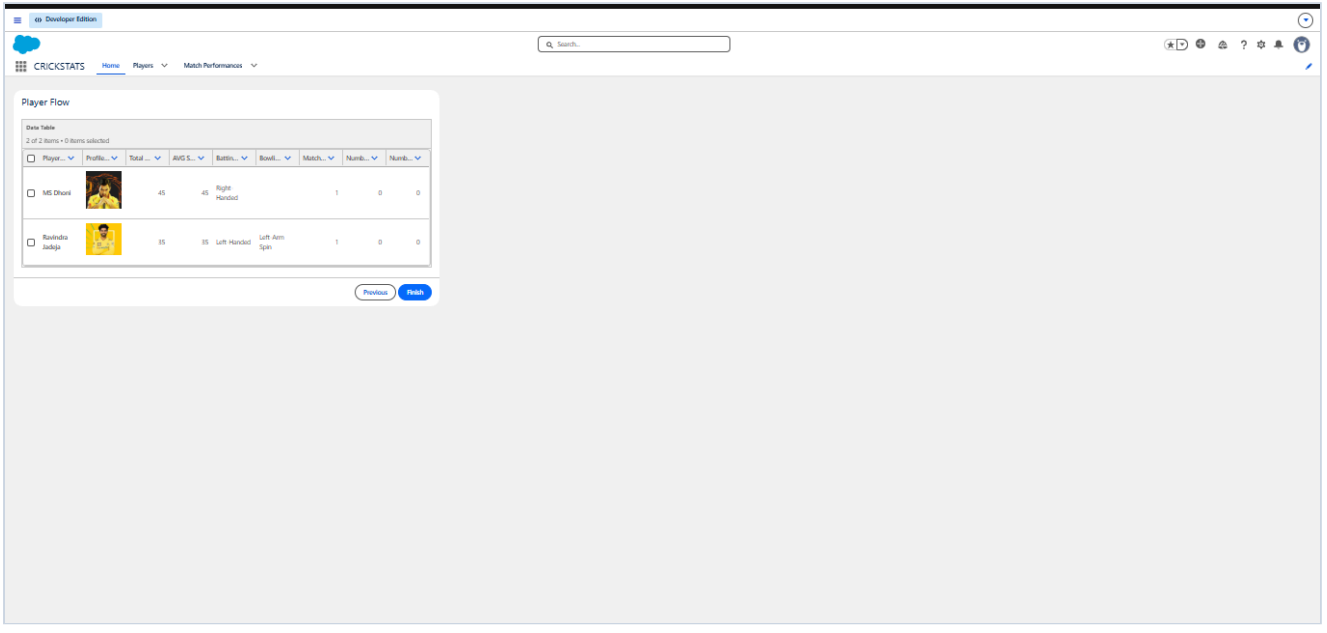
Sample Player Records



Sample Match Performance Records



Record	Player	Opponent	Result	Runs	Balls	Overs	Wickets
MP-0001	Virat Kohli	CSK	WON	82	55	-	-
MP-0002	MS Dhoni	MI	LOST	45	30	-	-
MP-0003	Jasprit Bumrah	RCB	WON	-	-	4	3
MP-0004	Rohit Sharma	CSK	WON	105	60	-	-
MP-0005	Ravindra Jadeja	MI	LOST	35	-	4	2



System Test Results

Test Scenarios	Expected Behavior	Result
Create Player records	Records save successfully without validation errors	**Passed**
Create Match Performance	Records save and link accurately to the parent player	**Passed**
Roll-up summary fields	Total Runs, Matches Played, etc., update automatically on save	**Passed**
AVG Score formula	Calculates accurately (Total Runs / Matches Played)	**Passed**
Player Flow validation	Filters correctly by Team (e.g., CSK shows Dhoni & Jadeja stats)	**Passed**

6. Advantages and Limitations

Advantages

- Centralized data for all players and matches
- Automated totals and average calculations
- Secure, cloud-based, and scalable
- Reduces manual effort and data entry errors

Limitations

- Requires an active Salesforce license
- Dependent on internet connectivity
- Limited offline access capabilities

7. Future Scope

- **Analytics:** Interactive reports and dashboards for deeper team/player insights.
- **Data Integrity:** Validation rules for score limits and mandatory fields.
- **Accessibility:** Mobile app integration for field managers.
- **Advanced Features:** AI-based performance prediction and automated player ranking systems.
- **Expansion:** Multi-league and tournament support architecture.

8. Conclusion

CRICKSTATS effectively demonstrates how the Salesforce platform can be leveraged to build a comprehensive data management and analytics application using a purely declarative (no-code) approach. By utilizing custom objects, Master-Detail relationships, roll-up summaries, formula fields, a Lightning app, and Screen Flows in tandem, the application provides quick, accurate, and scalable cricket performance insights.